

Reading structured data

Project: Python readiness portfolio

1. Prepare the purpose, prerequisites and materials

Teach the learning cycle

Primary target: Process a list of records consistently

Session · Sequence and minutes · Resources

Session: Session 1 · Understand · 45 min

Sequence and minutes: 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

Resources: Sections 1–6; record the brief recall in the workbook.

Session: Session 2 · Practise · 45 min

Sequence and minutes: 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

Resources: Sections 7–10; use the matching workbook W2 and misconception check.

Session: Session 3 · Investigate and transfer · 45 min

Sequence and minutes: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building. Repeated copies of a case share one recorded attempt; they are not new evidence.

Retrieval answer guidance

Dictionaries and records

Without opening the old lesson, explain one key idea from Dictionaries and records. Give an example and a mistake that the idea helps you avoid.

Look for: Read and update named dictionary fields

Reference example, not the only acceptable wording: Keys name, capacity, used; used is 2; remaining spaces 7.

Functions and arguments

Without opening the old lesson, explain one key idea from Functions and arguments. Give an example and a mistake that the idea helps you avoid.

Look for: Distinguish a function definition from a function call

Reference example, not the only acceptable wording: Name show_remaining; parameters capacity and used. Arguments 15 and 6 match capacity = 15 and used = 6. Expected printed result 9.

Branching and boundary tests

Without opening the old lesson, explain one key idea from Branching and boundary tests. Give an example and a mistake that the idea helps you avoid.

Look for: Trace the selected branch in an if elif else chain

Reference example, not the only acceptable wording: if tests negative first; elif checks ≤ 12 only after that fails; else handles the rest. Unindented print runs after any selected branch, whose assignment gives label a value.

Teach Week 11 as a two-session investigation using fictional data. Two fictional cupboards are stored as a list of dictionaries. Each record uses the same keys and units. We want the total spare spaces. Treat the records as a dataset, and trace one record at a time before trusting a total. Assess the three stated outcomes through the independent task, not by copying the worked result.

Outcomes

Process a list of records consistently

Compute a total from named fields

Preserve record meaning while tracing a loop

Preparation

Python 3 editor with a fresh local practice file

Paper trace table and the student workbook

Fictional input records from this lesson; no real student identifiers

Prerequisite answer

The key "used" stores 3.

End goal for the pictorial case

Trace reading structured data and explain the changed program before running it.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

2. Explain and model the mechanism

Two fictional cupboards are stored as a list of dictionaries. Each record uses the same keys and units. We want the total spare spaces. Treat the records as a dataset, and trace one record at a time before trusting a total.

The loop variable row refers to each record in turn. We calculate spare from that record, then add it. total starts before the loop. An empty list produces 0 because there are no updates; it does not tell us whether no cupboards exist or records were not collected.

Term · Meaning

Term: Dataset

Meaning: A collection of records

Term: Row

Meaning: One record in a table

Term: Field

Meaning: One named part of a record

Term: Accumulator

Meaning: A variable updated to keep a running result

Follow the evidence

Step or case · Evidence · Meaning

Step or case: First row

Evidence: $8 - 3 = 5$

Meaning: total becomes 5

Step or case: Second row

Evidence: $6 - 6 = 0$

Meaning: total stays 5

Read each row from left to right and connect the evidence to the stated rule.

Contributions 5, 0, 6; total 11. Exit: No. No records were processed; the result provides no evidence about actual cupboards.

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a prediction, then explain the deciding step. These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Add a third record with capacity 10 and used 4. Predict the total and list each contribution.

Thinking move · Teacher prompt

Thinking move: Plan

Teacher prompt: What is given? What is the question? Which procedure fits these facts?

Thinking move: Do

Teacher prompt: Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.

Thinking move: Check

Teacher prompt: Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Reading structured data

Read rows with a missing-field policy. A toy table is represented as a list of dictionaries; count only rows with a recorded count.

The facts we will use

Python 3 records: [{"count":2}, {"count":0}, {}]. Policy: include existing numeric counts, including zero; skip an absent field.

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Included values and absent field

	Count value	Include in mean?
Row 1	2	Yes
Row 2	0	Yes
Row 3	Absent	No

Read a cell using both its row and column labels.

Reading structured data

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Inspect three rows

Two rows contain count; the third does not.



2. Accumulate values

The included values sum to $2+0=2$, with denominator 2.



3. Report coverage and mean

Mean is 1; measured coverage is 2 of 3 rows.

CHANGE ONE THING

Add `{"count":4}` as a fourth row.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Inspect three rows

What happens and why: Two rows contain count; the third does not.

Step: 2. Accumulate values

What happens and why: The included values sum to $2+0=2$, with denominator 2.

Step: 3. Report coverage and mean

What happens and why: Mean is 1; measured coverage is 2 of 3 rows.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Structured data need a row policy before aggregation. Treating missing as zero changes the statistic and can conceal incomplete coverage; real zero remains an observation.

What this example does not establish

This worked row policy is explicit; file parsing, malformed values and data provenance require additional checks.

Change one thing in the pictured case

Change: Add {"count":4} as a fourth row.

Prediction: Sum becomes 6, denominator 3, mean 2; coverage is 3 of 4.

Reason: Only the added recorded value contributes; the existing absent field remains skipped.

Ask while showing the picture

Cover the result. Ask learners to name the inputs and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its evidence. Then change the supplied condition and compare. The picture is a representation; it does not document execution of a real AI model.

3. Demonstrate and guide practice

Time or stage · Teacher action

Time or stage: Session 1 start to 7 minutes

Teacher action: Retrieve the prerequisite and explain the purpose in the opening section.

Time or stage: 7 to 20 minutes

Teacher action: Read the worked example and diagram; ask learners to trace each step aloud.

Time or stage: 20 to 33 minutes

Teacher action: Complete W1 and W2, then compare predictions with observed results.

Time or stage: 33 to 45 minutes

Teacher action: Collect the guided prediction before running; discuss a mismatch and its cause.

Time or stage: Session 2 start to 8 minutes

Teacher action: Retrieve the main rule without showing the previous answer.

Time or stage: 8 to 20 minutes

Teacher action: Use W3 to diagnose and repair the stated failure; rerun an earlier case.

Time or stage: 20 to 35 minutes

Teacher action: Collect W4 independently, including code or plan, predictions and explanations, before hints.

Time or stage: 35 to 45 minutes

Teacher action: Discuss W5 and the exit; use the fresh retry when support was needed.

Use the worked example in Sections 3 and 6. Collect a prediction before revealing its result; ask the learner to point to each intermediate fact.

Lay out a row card beside the accumulator. Extension: count processed records separately and report that count with the total.

Reduce help deliberately

Completion task: For spare contributions 2, 4 and 1, trace total starting at 0.

Teacher check: Running totals 2, 6, 7.

Start by identifying the given inputs together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood. For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Reading structured data. Proposed explanation: "A successful run proves the conclusion". Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: This loop assumes every record has the required integer fields. A missing used field causes KeyError. Do not silently treat a missing count as zero; that would invent unused capacity.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class. If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Circle the included rows, calculate the sum and mean, and state coverage. Repeat after adding the fourth row with count four.

Reasoned teacher answer

Initially values 2 and 0 give sum 2 and mean 1 across two observations, coverage $2/3$. Adding 4 gives sum 6, mean 2 across three observations, coverage $3/4$.

Changed-case reasoning: Sum becomes 6, denominator 3, mean 2; coverage is 3 of 4. Only the added recorded value contributes; the existing absent field remains skipped.

Collect each learner's explanation before discussion. Use the first missing connection between input, deciding step and result to choose support. This question is worked-case practice, not fresh mastery evidence.

4. Run the investigation and diagnose misconceptions

A successful run proves the conclusion: This loop assumes every record has the required integer fields. A missing used field causes KeyError. Do not silently treat a missing count as zero; that would invent unused capacity.

Matching the example is enough: Require the new W4 case, explicit predictions and an explanation. Fresh retry: records (7, 2), (3, 3), (8, 6) contribute 5, 0, 2 and total 7; reversed order also totals 7.

A convenient shortcut preserves meaning: Putting `total = 0` inside the loop discards earlier contributions. With the original rows it finishes at 0, the second contribution. Move the initialization before the loop and rerun both the original and reversed row order.

Use Section 9 for the evidence ledger. Add a boundary or missing-input case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Match the support to the first error

What the response shows · Next teaching action

What the response shows: Missing or misread facts

Next teaching action: Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.

What the response shows: Wrong procedure

Next teaching action: Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.

What the response shows: Arithmetic or implementation error

Next teaching action: Trace intermediate values or lines; preserve the first mismatch and retry that step.

What the response shows: Correct result without a reason

Next teaching action: Ask for a trace and a changed case before marking independent understanding.

What the response shows: An unsupported wider claim

Next teaching action: Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal. Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

5. Assess and explain the answers

Guided and exit questions

Contributions 5, 0, 6; total 11. Exit: No. No records were processed; the result provides no evidence about actual cupboards.

W1 Read a row

In the worked dataset, state the number of records and the spare spaces in the second record.

There are 2 records. The second has 0 spare spaces.

W2 Trace all updates

For spare contributions 2, 4 and 1, trace total starting at 0.

Running totals 2, 6, 7.

W3 Repair initialization

A loop resets total to 0 on each pass over contributions 2, 4, 1. What final total appears and how should it be fixed?

It ends at 1. Initialize total once before the loop; repaired total 7.

W4 Process new records independently

Write a loop for records (capacity 9, used 2), (capacity 5, used 1), and (capacity 4, used 4). Predict the total. Reverse the order and test an empty list.

Contributions 7, 4, 0 give 11 in either order. Empty list gives 0. The loop must use both fields for every row.

W5 Interpret an empty result

An empty dataset produces total 0. Can you conclude that every cupboard is full?

No. No records were processed; the result provides no evidence about actual cupboards.

Independent assessment and fresh retry

Independent W4 evidence: Contributions 7, 4, 0 give 11 in either order. Empty list gives 0. The loop must use both fields for every row. Record whether the learner needed no help, a hint or explanation. Fresh retry: records (7, 2), (3, 3), (8, 6) contribute 5, 0, 2 and total 7; reversed order also totals 7.

Assess independent transfer

Selected case: Trace the named fields and missing-value policy. Predict the output, state the observed total and mean denominator, and report coverage. What changes if all three rows are empty records?

```
rows = [{"score": 2}, {"score": 6}, {}]
values = []
for row in rows:
    value = row.get("score")
    if value is not None:
        values.append(value)
observed_total = sum(values)
if len(values) > 0:
    mean = observed_total / len(values)
else:
    mean = None
print(observed_total, len(values), len(rows), mean)
```

Selected case answer

Output 8 2 3 4.0: observed total 8, two supplied scores out of three records, mean 4 over denominator 2.

With three empty records, output 0 0 3 None: the empty observed sum is 0 by the supplied summing procedure, coverage 0/3, and no observed mean exists. It does not claim the missing scores were zero or that the true total is known.

Dimension · Evidence required

Dimension: Explanation

Evidence required: Names the applicable idea or procedure and ties it to the supplied facts.

Dimension: Trace and result

Evidence required: Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.

Dimension: Limits

Evidence required: States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace. Do not penalise an honestly recorded model failure if the learner correctly explains its evidence and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case. Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

6. Connect the project, support and next lesson

Add a labelled reading structured data evidence sheet to Python readiness portfolio. Show how you can process a list of records consistently. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Lay out a row card beside the accumulator. Extension: count processed records separately and report that count with the total.

Next week: Testing and debugging functions. Reuse this week's explicit contracts and evidence.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

Plan the delayed return

Suggested return points in this level: python-bridge/12. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used. For a new level, revisit the previous capstone before checking the new prerequisites.

Record · Teacher evidence

Record: Learner code and date

Teacher evidence: _____

Record: Concept and case used; prior exposure

Teacher evidence: _____

Record: Explanation / trace / limits

Teacher evidence: _____

Record: Support and first repair

Teacher evidence: _____

Record: Changed case and delayed outcome

Teacher evidence: _____

Record: Next teaching action

Teacher evidence: _____

Use the evidence to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted outputs and the changed-case explanation. Use actual Python 3 results to check the prediction after reasoning first.

7. Prepare, teach and assess

Teaching target: Process a list of records consistently

Prepare the labelled worked-case picture, the matching workbook, fictional input cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

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Resources: Sections 1–6; record the brief recall in the workbook.

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Resources: Sections 7–10; use the matching workbook W2 and misconception check.

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Plan: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

Before class, solve the supported practice: For spare contributions 2, 4 and 1, trace total starting at 0.

Reasoned practice answer: Running totals 2, 6, 7.

Check every learner before discussion: Use the worked case for Reading structured data. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: This loop assumes every record has the required integer fields. A missing used field causes `KeyError`. Do not silently treat a missing count as zero; that would invent unused capacity.

Support: read the exact case aloud, enlarge the labelled picture, supply input cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Add {"count":4} as a fourth row. Require a prediction and a reason before comparing. Expected result: Sum becomes 6, denominator 3, mean 2; coverage is 3 of 4. Reason: Only the added recorded value contributes; the existing absent field remains skipped.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion · Not yet · With support · Independent

Criterion: Explanation

Not yet: Names an answer without a deciding fact

With support: Uses a hint to link fact and decision

Independent: Explains the deciding fact in own words

Criterion: Trace or test

Not yet: Cannot connect input and result

With support: Completes steps with prompts

Independent: Traces or tests a changed case accurately

Criterion: Limits and revision

Not yet: Claims more than evidence supports

With support: Names a limit after a prompt

Independent: States a defensible limit and useful next test

Project checkpoint: Python readiness portfolio. Keep runnable code, predictions, actual outputs, one repaired error and an explanation of a function on changed inputs.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and evidence.